



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22AI503/22AM503 MACHINE LEARNING

UNIT III – CLASSIFICATION

Naive Bayes Required Data Processing, M-Estimates, and Feature Selection

1. Introduction

1.1 Definition

While Naive Bayes is conceptually simple, building a production-quality NB classifier requires careful data processing: text normalisation, handling the zero-frequency problem via smoothing (Laplace and M-estimates), and selecting an informative, non-redundant feature subset. These steps directly determine whether the independence assumption and probability estimates remain sound in practice.

Definition:

Required Data Processing: domain-specific steps (tokenization, discretization, vectorisation) to prepare raw data for NB's probability estimation.

M-Estimate: a generalised smoothing technique that blends observed counts with a prior probability, controlled by an equivalent sample size parameter m .

Feature Selection for NB: removing irrelevant or redundant features improves both probability estimate quality and computational efficiency.

1.2 Importance in Machine Learning

Unlike distance-based models, NB does not need feature scaling — but it is highly sensitive to how categorical/text data is processed and how zero counts are handled. Poor preprocessing (no smoothing, no stop-word removal, no feature pruning) can make an otherwise sound NB model fail catastrophically or perform far below its potential.

1.3 Role in the Classification Pipeline

This stage sits between general data preprocessing (Unit II) and model training: text/categorical data is vectorised into the count or presence representation NB expects, smoothing parameters are chosen, and a feature subset is selected before the model is fit. These choices are often as important to final accuracy as the choice of NB variant itself.

2. Learning Outcomes

Understand

- The specific data processing requirements for Naive Bayes (text and categorical pipelines)
- The zero-frequency problem and why it specifically threatens NB's multiplicative formula
- The M-estimate formula and its relationship to Laplace smoothing

Apply

- Build a text preprocessing pipeline (tokenize, stop-word removal, stemming, vectorise)
- Implement Laplace and M-estimate smoothing manually and via scikit-learn parameters
- Apply Chi-squared and Mutual Information feature selection for NB text classifiers

Analyze

- Evaluate the effect of smoothing strength (m) on model bias and variance
- Compare Laplace smoothing against domain-informed M-estimates for small training sets
- Diagnose when feature selection improves vs. harms NB performance

3. Detailed Technical Concepts

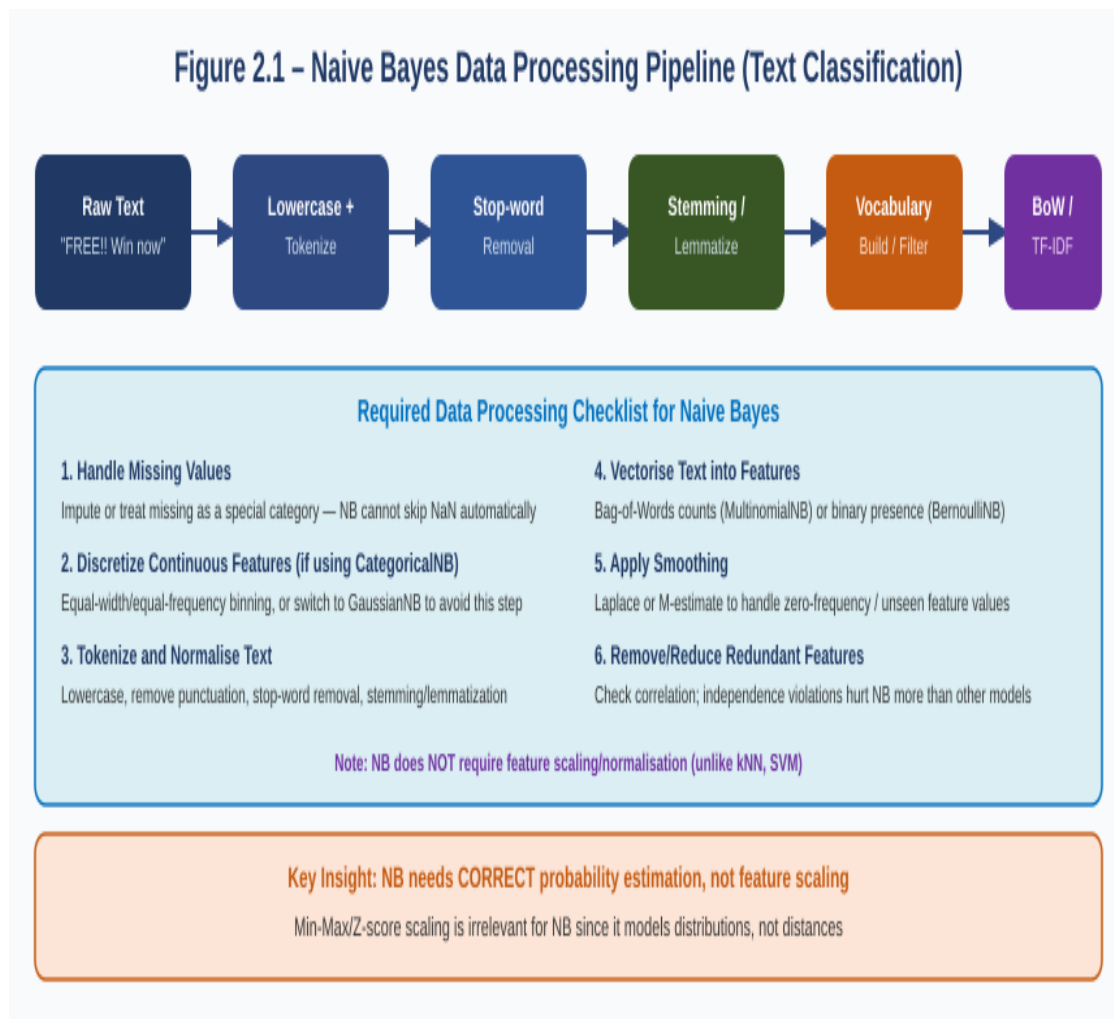


Figure 2.1 – Naive Bayes Data Processing Pipeline (Text Classification)

3.1 Required Data Processing for Naive Bayes

3.1.1 Handling Missing Values

Unlike some algorithms, NB cannot natively skip missing values during probability estimation — each missing entry must be either imputed (mode for categorical, mean/median for numeric) or treated as its own explicit category ("Missing"), which can sometimes itself be informative (e.g., a survey question left blank may correlate with the target).

3.1.2 Discretizing Continuous Features (for Categorical/Multinomial NB)

If using CategoricalNB or building a frequency-table-based classifier (the classical textbook approach), continuous attributes must first be discretized into bins (equal-width, equal-frequency, or entropy-based — see Unit II Topic 6). Alternatively, switching to GaussianNB avoids this step entirely by modelling the continuous distribution directly.

3.1.3 Text Preprocessing Pipeline

- Lowercasing: normalise case so "Free" and "free" are treated as the same token.
- Tokenization: split text into individual word/token units.
- Stop-word Removal: remove highly frequent, low-information words ("the", "is", "and") that add noise without discriminative value.
- Stemming/Lemmatization: reduce words to root form ("running"→"run") to consolidate sparse counts across variants of the same word.
- Punctuation/Special Character Handling: strip or selectively retain punctuation that carries signal (e.g., "!!!" in spam detection).

3.1.4 Vectorisation

Convert processed tokens into numeric feature vectors:

- Bag-of-Words (BoW) counts: feeds MultinomialNB. sklearn: CountVectorizer().
- Binary presence/absence: feeds BernoulliNB. sklearn: CountVectorizer(binary=True).
- TF-IDF weighting: down-weights common terms, up-weights distinctive terms — often combined with MultinomialNB via TfidfVectorizer() for improved performance over raw counts.

3.1.5 What NB Does NOT Require

Unlike kNN or SVM, Naive Bayes does NOT require feature scaling/normalisation (Min-Max, Z-score) because it models per-feature probability distributions rather than distances — the scale of one feature does not influence the likelihood computation of another.

3.2 The Zero-Frequency Problem (Detailed)

When a feature value never co-occurs with a class in the training data, the raw maximum-likelihood estimate $P(x_i|Y)=0$. Because NB multiplies likelihoods across all features, a single zero term forces the entire posterior score to zero — regardless of how strongly every other feature points to that class.

This is especially common with: rare words in text classification, small training sets, high-cardinality categorical features, and test-time feature values never seen during training.

3.3 Smoothing Techniques

3.3.1 Laplace (Add-One) Smoothing

$P(x_i|Y) = (\text{count}(x_i, Y) + 1) / (\text{count}(Y) + k)$ where k = number of possible values for feature x_i .

Adds a pseudo-count of 1 to every possible feature value, ensuring no probability is ever exactly zero. Simple, parameter-free (aside from the implicit assumption of a uniform prior over feature values), and the default in most software (sklearn $\alpha=1.0$).

3.3.2 M-Estimate (Generalised Smoothing)

$P(x_i|Y) = (\text{count}(x_i, Y) + m \times p) / (\text{count}(Y) + m)$

- m : "equivalent sample size" — controls how strongly the prior p influences the estimate relative to observed data. Larger m = more smoothing, more reliance on the prior.
- p : the prior probability estimate for the feature value — can be set to a domain-informed value (e.g., known background rate) rather than assuming uniform $1/k$ as Laplace does.

Laplace smoothing is the special case where $m=k$ and $p=1/k$ (uniform prior). M-estimates generalise this by allowing both the smoothing strength and the prior belief to be tuned independently — valuable when domain knowledge about expected feature distributions exists, or when k is large and uniform-prior Laplace over-smooths.

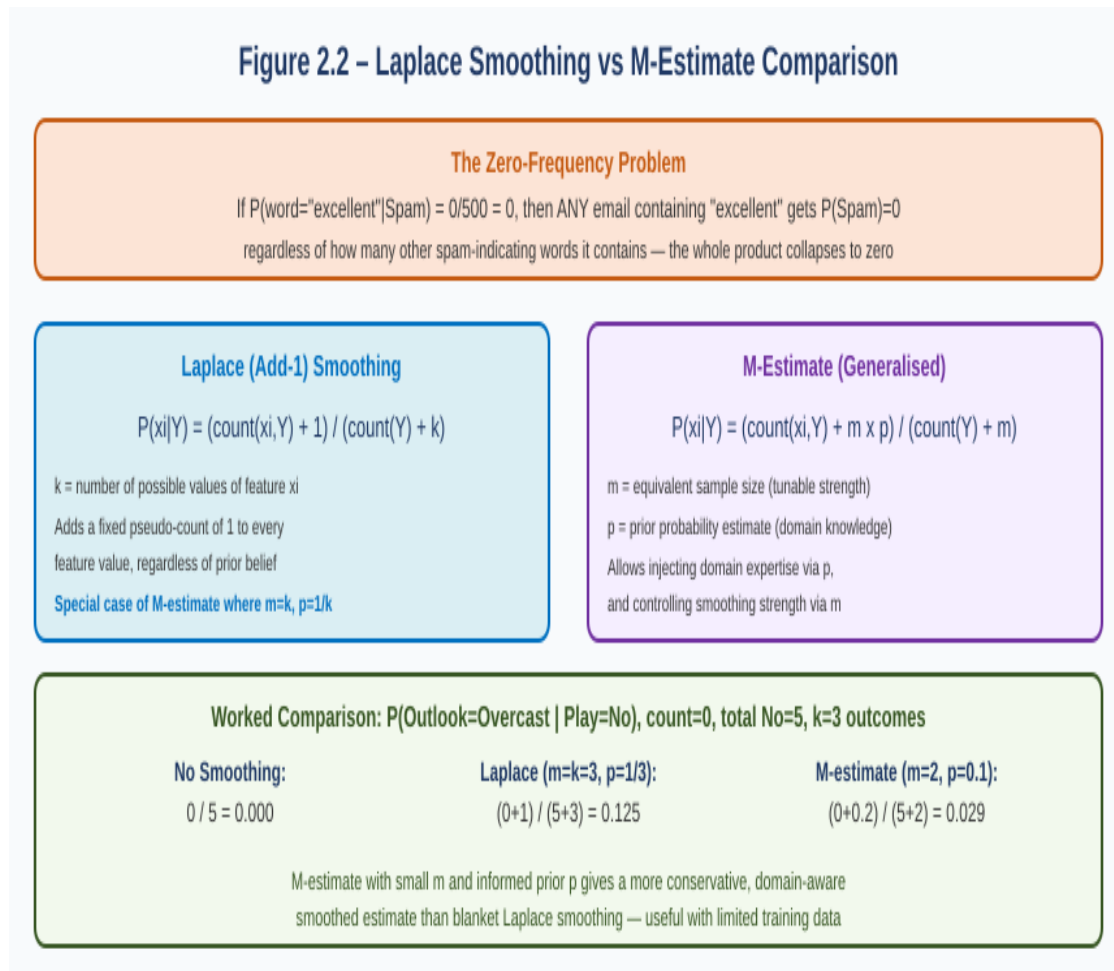


Figure 2.2 – Laplace Smoothing vs M-Estimate: Formula Comparison and Worked Example

3.3.3 Choosing the Smoothing Strength

Small m : estimates stay close to observed counts (low bias, high variance — risky with sparse data). Large m : estimates pull strongly toward the prior p (high bias, low variance — safer with very limited data, but may mask real signal). The optimal m is typically tuned via cross-validation, especially in domains with limited training examples per class.

In scikit-learn, the α parameter in `MultinomialNB(alpha=1.0)` and `BernoulliNB(alpha=1.0)` implements Laplace/Lidstone smoothing; $\alpha < 1$ is "Lidstone smoothing" (less aggressive than Laplace), and $\alpha=0$ disables smoothing (not recommended).

3.4 Feature Selection for Naive Bayes

Naive Bayes benefits substantially from feature selection — removing irrelevant or redundant features both reduces noise in the cumulative log-sum and lessens independence-assumption violations from correlated features.

Figure 2.3 – Feature Selection Methods for Naive Bayes

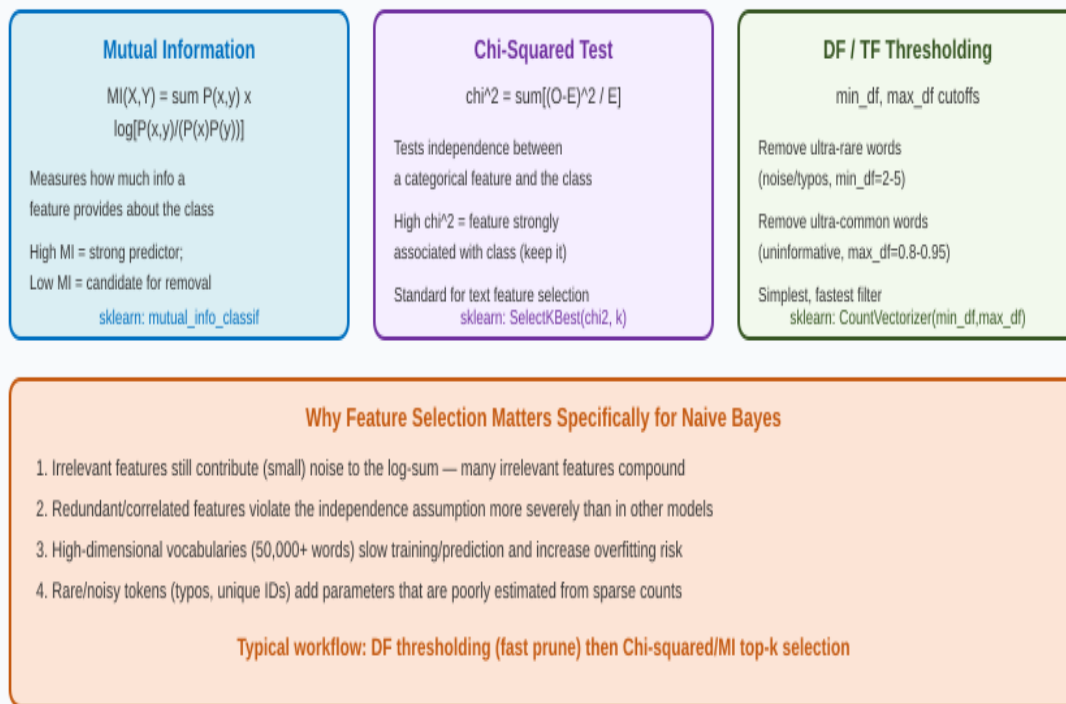


Figure 2.3 – Feature Selection Methods for Naive Bayes: Mutual Information, Chi-Squared, DF Thresholding

3.4.1 Mutual Information

$MI(X,Y) = \sum \text{over } x,y \text{ of } P(x,y) \times \log\left[\frac{P(x,y)}{P(x)P(y)}\right]$. Measures how much knowing the feature value reduces uncertainty about the class. Naturally suited to NB since it is grounded in the same probabilistic framework. sklearn: `mutual_info_classif(X, y)`.

3.4.2 Chi-Squared (χ^2) Test

$\chi^2 = \sum \left[\frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} \right]$. Tests whether a categorical feature is statistically independent of the class label. High chi-squared scores indicate strong class association — the standard feature-selection method for text classification (selecting the top-k most class-informative words). sklearn: `SelectKBest(chi2, k=2000)`.

3.4.3 Document Frequency / Term Frequency Thresholding

A fast, simple pre-filter: remove terms appearing in too few documents (min_df, e.g., < 2-5 documents — likely noise, typos, or overly rare tokens) and terms appearing in too many documents (max_df, e.g., > 80-95% of documents — uninformative, near-stop-words). Applied before or alongside statistical feature selection. sklearn: `CountVectorizer(min_df=5, max_df=0.9)`.

3.5 Recommended Workflow

1) Apply DF thresholding for a fast initial prune of the vocabulary. 2) Apply Chi-squared or Mutual Information to select the top-k most informative remaining features. 3) Check for and remove highly correlated/redundant derived features. 4) Apply smoothing (Laplace as default; M-estimate with informed priors when domain knowledge is available or training data is very limited). 5) Validate via cross-validation, tuning k (feature count) and alpha/m (smoothing strength) jointly.

4. Architecture / Processing and Smoothing Workflow

Step	Action	Tool / Method	Output
1	Clean and tokenize text/categories	Lowercase, regex, NLTK/spaCy tokenizer	Cleaned token list per document
2	Remove stop-words, stem/lemmatize	NLTK stopwords, PorterStemmer	Reduced, normalised token list
3	Build vocabulary with DF thresholds	CountVectorizer(min_df, max_df)	Pruned vocabulary
4	Vectorise into BoW / TF-IDF / binary	CountVectorizer / TfidfVectorizer	Feature matrix X
5	Select top-k features	SelectKBest(chi2 or mutual_info)	Reduced feature matrix
6	Fit NB with smoothing parameter	MultinomialNB(alpha=1.0)	Trained model with smoothed probabilities
7	Tune alpha/m via cross-validation	GridSearchCV over alpha values	Optimal smoothing strength

5. Industry Tools and Technologies

5.1 Commercial Tools

Tool	Vendor	Smoothing / Feature Selection Support
RapidMiner	RapidMiner	Naive Bayes operator with Laplace correction checkbox
SAS Text Miner	SAS	Built-in chi-squared and TF-IDF feature reduction
IBM Watson NLU	IBM	Managed text preprocessing pipeline, custom classifiers
Google Cloud NL API	Google	Pre-trained classification with internal smoothing
MonkeyLearn	MonkeyLearn	No-code text classifier with NB backend

5.2 Open-Source Tools

- scikit-learn: CountVectorizer, TfidfVectorizer, SelectKBest(chi2/mutual_info_classif), MultinomialNB(alpha=)
- NLTK: word_tokenize, stopwords corpus, PorterStemmer, WordNetLemmatizer
- spaCy: industrial-strength tokenization, lemmatization, and stop-word lists
- gensim: Dictionary filtering (filter_extremes for DF thresholding), TF-IDF models
- e1071 (R): naiveBayes(laplace=1) parameter for smoothing strength

6. Real-World Applications

Domain	Processing Challenge	Smoothing/Selection Approach	Result
Spam Filtering	New/rare spam vocabulary (zero-freq)	Laplace alpha=1, daily vocab refresh	Adapts to evolving spam tactics

Medical Diagnosis	Rare symptom-disease combos	M-estimate with clinical prior p	Avoids overconfident zero-probability misdiagnosis
Legal Document Tagging	50,000+ word legal vocabulary	Chi-squared top-5000 features	10x faster training, comparable accuracy
Customer Support Routing	Ticket text with typos/jargon	min_df=3 + lemmatization	Reduced noise, improved routing accuracy
Fraud Pattern Detection	Imbalanced rare fraud categories	M-estimate with low prior p for fraud	Better calibrated probability for rare class

7. Advantages and Limitations

Advantages	Limitations
Smoothing prevents catastrophic zero-probability failures	Choosing optimal smoothing strength (m/alpha) requires tuning
M-estimates allow injecting valuable domain knowledge via prior p	Over-smoothing (large m) can wash out genuine rare-but-real signals
Feature selection reduces noise and computational cost significantly	Aggressive feature pruning risks removing rare but highly discriminative terms
NB requires no feature scaling, simplifying the pipeline	Text preprocessing choices (stemming vs lemmatization) can meaningfully shift results
DF thresholding is extremely fast even on huge vocabularies	Chi-squared/MI selection adds computational overhead for very large feature sets

8. Case Study

CASE STUDY

Case: 20 Newsgroups Text Classification — Smoothing and Feature Selection Impact

Problem: A research team benchmarking text classifiers on the classic 20 Newsgroups dataset (20 categories, ~18,000 documents) found baseline MultinomialNB accuracy of only 71% without any tuning.

Dataset: 18,846 newsgroup posts, vocabulary of 130,000+ raw tokens after basic tokenization.

Investigation Phase 1 — Smoothing: Without smoothing ($\alpha=0$): many test documents received $P=0$ for the correct class due to unseen words. With default Laplace ($\alpha=1.0$): accuracy jumped to 82%. Grid search over α in [0.001, 0.01, 0.1, 0.5, 1.0, 2.0] found $\alpha=0.1$ (Lidstone, lighter than Laplace) optimal: accuracy 85%.

Investigation Phase 2 — Feature Selection: Raw 130,000-feature vocabulary included many noisy tokens (email headers, signatures, rare misspellings). Applied min_df=3, max_df=0.9 DF thresholding: vocabulary reduced to 35,000. Applied chi-squared SelectKBest($k=10000$): final vocabulary 10,000 highly class-informative words.

Combined Result: $\alpha=0.1$ + chi-squared top-10K features: accuracy = 89.3%, training time reduced from 4.2s to 0.8s, prediction latency reduced 6x.

Conclusion: Systematic smoothing tuning (+14 points) and feature selection (+4.3 points, plus large efficiency gains) together transformed a mediocre baseline into a strong, fast, production-ready text classifier — without changing the underlying NB algorithm at all.

9. Scenario-Based Learning

SCENARIO 1

SCENARIO 1: Unseen Word Crashes Production Spam Filter

Background: A startup's spam filter, trained on 6 months of historical email, suddenly starts misclassifying a wave of new spam containing the novel term "cryptoairdrop" — a word never seen during training.

Technical Environment: Python, scikit-learn MultinomialNB(alpha=1.0), daily retraining pipeline.

Problem Statement: Despite using Laplace smoothing (alpha=1.0), the team observes spam recall dropping from 96% to 71% over a 2-week period as new crypto-spam vocabulary emerges.

Available Evidence: Vocabulary is fixed at training time via CountVectorizer.fit() on historical data; new words at prediction time are simply ignored (out-of-vocabulary), losing their signal entirely — smoothing only helps for IN-vocabulary words with zero class-count, not OOV words.

Investigation: Distinguish between "in-vocabulary, zero-count-per-class" (handled by smoothing) versus "out-of-vocabulary entirely" (not handled by smoothing at all — silently dropped).

Analysis: The real fix is not more smoothing but more frequent retraining/revectorization to capture emerging vocabulary, combined with a fallback mechanism for OOV-heavy messages.

Findings: Switching to daily vocabulary rebuilding (retrain CountVectorizer + model nightly on rolling 30-day window) restored spam recall to 95% within 3 days of the new spam wave appearing.

Recommended Solution: Smoothing solves the zero-COUNT problem for known vocabulary; it does NOT solve the out-of-vocabulary problem. Production NB text systems need a vocabulary refresh/retraining cadence matched to how quickly the domain vocabulary evolves.

SCENARIO 2

SCENARIO 2: M-Estimate with Domain Prior for Rare Disease Diagnosis

Background: A hospital builds a NB-based diagnostic support tool for a rare disease (prevalence 0.3% in the population) using symptom presence/absence features from only 800 historical patient records (240 positive cases due to oversampling for training).

Technical Environment: Python, custom NB implementation, CategoricalNB-style frequency tables, 15 symptom features.

Problem Statement: Using default Laplace smoothing (uniform prior 1/k) on the small training set produces poorly calibrated probabilities — the model recommends unnecessary follow-up tests for many low-risk patients because smoothing pulls symptom likelihoods toward an uninformative 50/50 prior.

Available Evidence: With only 800 records, several symptom-disease combinations have very low counts (5-10 instances), and Laplace smoothing's uniform prior assumption does not reflect the true rarity of the disease.

Investigation: Replace Laplace ($m=k$, $p=1/k$) with M-estimate using a clinically-informed prior p , derived from published epidemiological base rates for each symptom given the disease, and a moderate m ($m=10$) to balance prior knowledge against the (limited) observed data.

Analysis: M-estimate formula: $P(\text{symptom}|\text{disease}) = (\text{count} + m \cdot p_{\text{clinical}}) / (\text{count}_{\text{disease}} + m)$. This anchors estimates to known medical literature rather than an uninformative uniform assumption.

Findings: False-positive referral rate dropped from 34% to 12% with M-estimate + clinical priors, while maintaining the same 91% sensitivity (true positive rate) for the rare disease.

Recommended Solution: When training data is limited and domain expertise (clinical literature, expert priors) is available, M-estimate smoothing with an informed prior p substantially outperforms generic Laplace smoothing, especially for rare-class problems.

SCENARIO 3

SCENARIO 3: Over-Aggressive Feature Selection Removes Key Signal

Background: A legal-tech company builds a contract-clause classifier (10 clause types) using MultinomialNB with chi-squared feature selection. Initial vocabulary: 40,000 legal terms.

Technical Environment: Python, scikit-learn, SelectKBest(chi2), 25,000 labelled contract clauses.

Problem Statement: To speed up training, the team aggressively reduced features to the top 200 by chi-squared score. Accuracy on the "Force Majeure" clause type dropped from 94% to 61%, while other clause types stayed roughly stable.

Available Evidence: The phrase "act of God" — a near-perfect indicator of Force Majeure clauses — was DF-thresholded out earlier ($\text{min_df}=10$) because it appears in only 6 documents total (this clause type is rare in the training set), so it never reached the chi-squared ranking stage at all.

Investigation: Review DF thresholding settings against per-class document counts — $\text{min_df}=10$ silently removes any term appearing in fewer than 10 documents, disproportionately hurting rare classes/rare-but-decisive terms.

Analysis: Aggressive global thresholds (min_df , top-k) do not account for per-class rarity — a term can be globally rare yet a near-perfect signal for one specific class.

Findings: Lowering min_df to 2 and increasing top-k to 2,000 restored "act of God" and similar rare-but-decisive phrases to the vocabulary. Force Majeure accuracy recovered to 93%; overall accuracy improved from 84% to 91%.

Recommended Solution: When classes are imbalanced or some classes are rare, validate feature selection thresholds per-class (not just globally) and prefer slightly more conservative DF thresholds combined with class-balanced chi-squared scoring to avoid losing rare-but-critical features.

10. Common Errors and Troubleshooting

Error	Cause	Impact	Fix
Disabling smoothing ($\alpha=0$)	Misunderstanding default parameters	Zero-frequency problem causes wrong/undefined predictions	Always keep $\alpha>0$; default to Laplace ($\alpha=1.0$)
Assuming smoothing fixes OOV words	Confusing in-vocab zero-count with out-of-vocab	Persistent silent accuracy decay over time	Implement periodic vocabulary/model retraining
Uniform Laplace prior on imbalanced rare classes	Not using M-estimate with informed prior	Poorly calibrated probabilities for rare classes	Use M-estimate with domain-informed prior p
Global DF thresholds on imbalanced data	Ignoring per-class term rarity	Loses rare-but-decisive features for minority classes	Lower min_df or apply per-class-aware feature selection
Scaling/normalising NB input features	Habit carried over from kNN/SVM pipelines	Unnecessary processing step (NB does not need scaling)	Skip scaling for NB; focus effort on smoothing and feature selection instead

11. Key Terminologies

Term	Definition
Zero-Frequency Problem	When an unseen feature-class combination yields $P=0$, collapsing the entire posterior product
Laplace (Add-1) Smoothing	Adds a pseudo-count of 1 to every feature value to avoid zero probabilities
M-Estimate	Generalised smoothing: $(\text{count} + m \cdot p) / (\text{total} + m)$, blending observed data with a tunable prior
Equivalent Sample Size (m)	Parameter controlling how strongly the prior p influences the M-estimate

Prior Probability (p)	Domain-informed expected probability used in M-estimate smoothing
Lidstone Smoothing	Generalisation of Laplace using $\alpha < 1$ for lighter smoothing
Stop-words	Highly frequent, low-information words removed during text preprocessing
Stemming / Lemmatization	Reducing words to root/base form to consolidate sparse token counts
Bag-of-Words (BoW)	Text representation as word frequency counts, ignoring order
TF-IDF	Term Frequency-Inverse Document Frequency weighting scheme for text features
Mutual Information (MI)	Measure of how much a feature reduces uncertainty about the class
Chi-Squared Feature Selection	Statistical test selecting features most associated with the class label
DF Thresholding (min_df/max_df)	Pruning vocabulary based on document frequency bounds
Out-of-Vocabulary (OOV)	A term seen at prediction time that was absent from the training vocabulary

12. Topic Summary

UNIT III – Topic 2

NAIVE BAYES — DATA PROCESSING, M-ESTIMATES, FEATURE SELECTION — SUMMARY

- Required processing: handle missing values, discretize if needed, full text pipeline (tokenize, stop-words, stem/lemmatize, vectorise). NB does NOT need feature scaling.
- Zero-frequency problem: an unseen feature-class pair forces the entire posterior product to zero — must be addressed via smoothing.
- Laplace smoothing: $P(x_i|Y) = (\text{count} + 1) / (\text{total} + k)$, uniform prior, simple default.
- M-estimate: $P(x_i|Y) = (\text{count} + m \cdot p) / (\text{total} + m)$, allows domain-informed prior p and tunable strength m — superior with small data or known base rates.
- Smoothing fixes in-vocabulary zero counts, but NOT out-of-vocabulary words at prediction time — that requires periodic retraining.
- Feature selection: Mutual Information and Chi-squared rank features by class-relevance; DF thresholding (min_df/max_df) provides a fast initial prune.
- Caution: aggressive global thresholds can remove rare-but-decisive features, especially for minority classes — validate per-class impact.
- Tools: scikit-learn (CountVectorizer, TfidfVectorizer, SelectKBest, MultinomialNB alpha parameter), NLTK, spaCy, gensim.